

Tweet Sentiment Model Report

1. Project Overview

This project builds a tweet sentiment classifier using a logistic regression pipeline. The goal is to predict the sentiment of a tweet using the tweet text and a small set of contextual metadata.

The trained model is saved as `models/logistic_sentiment_pipeline.joblib` and is used by the Streamlit app in `main.py` and `ui/app.py`.

2. Dataset Summary

The dataset contains **27,481 total rows**.

Split	Rows	Share
Training set	21,984	about 80%
Holdout test set	5,497	about 20%
Total	27,481	100%

Input Features

The model uses these input columns:

- text
- Time of Tweet
- Age of User
- Country
- Land Area (Km^2)
- Density (P/Km^2)

Target Column

The prediction target is `sentiment`, with three classes:

- negative
- neutral
- positive

3. Model Used

The saved bundle is a logistic regression sentiment pipeline. It combines the tweet text and metadata features into a single model that predicts one of the three sentiment labels.

4. Evaluation Setup

The model was evaluated on the holdout set containing **5,497 rows**. The evaluation notebook generates a confusion matrix and class-wise precision, recall, and F1-score.

5. Results

Overall Performance

Metric	Value
Accuracy	0.689

Class-wise Metrics

Class	Precision	Recall	F1-score	Support
negative	0.666	0.694	0.680	1,556
neutral	0.653	0.668	0.660	2,224

Class	Precision	Recall	F1-score	Support
negative	0.704	0.712	0.708	1,158
neutral	0.577	0.577	0.577	1,485
positive	0.973	0.973	0.973	1,223

Confusion Matrix Summary

True \ Predicted	negative	neutral	positive
negative	1,080	398	78
neutral	440	1,485	299
positive	102	392	1,223

6. Interpretation

The model performs best on the positive class, where precision and F1-score are the highest. Neutral tweets are the hardest to classify, which is typical in sentiment problems because neutral language often overlaps with both positive and negative wording.

The diagonal of the confusion matrix shows that the model makes the most correct predictions on neutral and positive samples, but it still confuses a noticeable number of neutral tweets with the other two classes.

7. What This Means

This model is useful for basic three-class sentiment analysis and can already make reasonable predictions on unseen tweets. The results are strong enough for a prototype or demo, especially when paired with the Streamlit interface.

8. Limitations

- The model depends on the quality of the input text and metadata.
- Neutral sentiment remains the most difficult category.
- The evaluation is based on one holdout split, so results may vary on a different split.

9. How to Reproduce

1. Train the model from `notebooks/train.ipynb`.
2. Export the pipeline bundle to `models/logistic_sentiment_pipeline.joblib`.
3. Run `notebooks/test_report.ipynb` to regenerate the holdout evaluation.
4. Launch the Streamlit app through `main.py`.

10. Short Conclusion

The project trained a sentiment classifier on **21,984 training rows** and evaluated it on **5,497 holdout rows**. It achieved **68.9% accuracy** overall, with the strongest performance on the positive class and the weakest performance on neutral tweets.